

Using the definition of multiplication modulo n , we define the **multiplicative group modulo n** as $(\mathbb{Z}_n^*, \cdot_n)$. The elements of this group are the set $\mathbb{Z}_n^*$ of elements in $\mathbb{Z}_n$ that are relatively prime to n , so that each one has a unique inverse, modulo n :

$$\mathbb{Z}_n^* = \{[a]_n \in \mathbb{Z}_n : \gcd(a, n) = 1\} .$$

To see that $\mathbb{Z}_n^*$ is well defined, note that for $0 \leq a < n$, we have $a = (a + kn) \pmod{n}$ for all integers k . By Exercise 31.2-3, therefore, $\gcd(a, n) = 1$ implies $\gcd(a + kn, n) = 1$ for all integers k . Since $[a]_n = \{a + kn : k \in \mathbb{Z}\}$, the set $\mathbb{Z}_n^*$ is well defined. An example of such a group is

$$\mathbb{Z}_{15}^* = \{1, 2, 4, 7, 8, 11, 13, 14\} ,$$

where the group operation is multiplication modulo 15. (We have denoted an element $[a]_{15}$ as a , and thus, for example, we denote $[7]_{15}$ as 7.) Figure 31.2(b) shows the group $(\mathbb{Z}_{15}^*, \cdot_{15})$. For example, $8 \cdot 11 = 13 \pmod{15}$, working in $\mathbb{Z}_{15}^*$. The identity for this group is 1.

Theorem 31.13

The system $(\mathbb{Z}_n^*, \cdot_n)$ is a finite abelian group.

Proof Theorem 31.6 implies that $(\mathbb{Z}_n^*, \cdot_n)$ is closed. Associativity and commutativity can be proved for $\cdot_n$ as they were for $+_n$ in the proof of Theorem 31.12. The identity element is $[1]_n$. To show the existence of inverses, let a be an element of $\mathbb{Z}_n^*$ and let (d, x, y) be returned by EXTENDED-EUCLID(a, n). Then we have $d = 1$, since $a \in \mathbb{Z}_n^*$, and

$$ax + ny = 1 , \tag{31.20}$$

or equivalently,

$$ax = 1 \pmod{n} .$$

Thus $[x]_n$ is a multiplicative inverse of $[a]_n$, modulo n . Furthermore, we claim that $[x]_n \in \mathbb{Z}_n^*$. To see why, equation (31.20) demonstrates that the smallest positive linear combination of x and n must be 1. Therefore, Theorem 31.2 implies that $\gcd(x, n) = 1$. We defer the proof that inverses are uniquely defined until Corollary 31.26 in Section 31.4. ■

As an example of computing multiplicative inverses, suppose that $a = 5$ and $n = 11$. Then EXTENDED-EUCLID(a, n) returns $(d, x, y) = (1, -2, 1)$, so that $1 = 5 \cdot (-2) + 11 \cdot 1$. Thus, $[-2]_{11}$ (i.e., $[9]_{11}$) is the multiplicative inverse of $[5]_{11}$.

When working with the groups $(\mathbb{Z}_n, +_n)$ and $(\mathbb{Z}_n^*, \cdot_n)$ in the remainder of this chapter, we follow the convenient practice of denoting equivalence classes by their representative elements and denoting the operations $+_n$ and $\cdot_n$ by the usual

arithmetic notations $+$ and $\cdot$ (or juxtaposition, so that $ab = a \cdot b$) respectively. Furthermore, equivalences modulo n may also be interpreted as equations in $\mathbb{Z}_n$. For example, the following two statements are equivalent:

$$ax = b \pmod{n}$$

and

$$[a]_n \cdot_n [x]_n = [b]_n .$$

As a further convenience, we sometimes refer to a group $(S, \oplus)$ merely as S when the operation $\oplus$ is understood from context. We may thus refer to the groups $(\mathbb{Z}_n, +_n)$ and $(\mathbb{Z}_n^*, \cdot_n)$ as just $\mathbb{Z}_n$ and $\mathbb{Z}_n^*$, respectively.

We denote the (multiplicative) inverse of an element a by $(a^{-1} \pmod{n})$. Division in $\mathbb{Z}_n^*$ is defined by the equation $a/b = ab^{-1} \pmod{n}$. For example, in $\mathbb{Z}_{15}^*$ we have that $7^{-1} = 13 \pmod{15}$, since $7 \cdot 13 = 91 = 1 \pmod{15}$, so that $2/7 = 2 \cdot 13 = 11 \pmod{15}$.

The size of $\mathbb{Z}_n^*$ is denoted $\phi(n)$. This function, known as *Euler's phi function*, satisfies the equation

$$\phi(n) = n \prod_{p \text{ prime such that } p \mid n} \left(1 - \frac{1}{p}\right) , \quad (31.21)$$

so that p runs over all the primes dividing n (including n itself, if n is prime). We won't prove this formula here. Intuitively, begin with a list of the n remainders $\{0, 1, \dots, n-1\}$ and then, for each prime p that divides n , cross out every multiple of p in the list. For example, since the prime divisors of 45 are 3 and 5,

$$\begin{aligned} \phi(45) &= 45 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) \\ &= 45 \left(\frac{2}{3}\right) \left(\frac{4}{5}\right) \\ &= 24 . \end{aligned}$$

If p is prime, then $\mathbb{Z}_p^* = \{1, 2, \dots, p-1\}$, and

$$\begin{aligned} \phi(p) &= p \left(1 - \frac{1}{p}\right) \\ &= p - 1 . \end{aligned} \quad (31.22)$$

If n is composite, then $\phi(n) < n - 1$, although it can be shown that

$$\phi(n) > \frac{n}{e^\gamma \ln \ln n + 3/\ln \ln n} \quad (31.23)$$

for $n \geq 3$, where $\gamma = 0.5772156649\dots$ is *Euler's constant*. A somewhat simpler (but looser) lower bound for $n > 5$ is

$$\phi(n) > \frac{n}{6 \ln \ln n} . \quad (31.24)$$

The lower bound (31.23) is essentially the best possible, since

$$\liminf_{n \rightarrow \infty} \frac{\phi(n)}{n / \ln \ln n} = e^{-\gamma} . \quad (31.25)$$

Subgroups

If $(S, \oplus)$ is a group, $S' \subseteq S$, and $(S', \oplus)$ is also a group, then $(S', \oplus)$ is a *subgroup* of $(S, \oplus)$. For example, the even integers form a subgroup of the integers under the operation of addition. The following theorem, whose proof we leave as Exercise 31.3-3, provides a useful tool for recognizing subgroups.

Theorem 31.14 (A nonempty closed subset of a finite group is a subgroup)

If $(S, \oplus)$ is a finite group and S' is any nonempty subset of S such that $a \oplus b \in S'$ for all $a, b \in S'$, then $(S', \oplus)$ is a subgroup of $(S, \oplus)$. ■

For example, the set $\{0, 2, 4, 6\}$ forms a subgroup of $\mathbb{Z}_8$, since it is nonempty and closed under the operation $+$ (that is, it is closed under $+_8$).

The following theorem, whose proof is omitted, provides an extremely useful constraint on the size of a subgroup.

Theorem 31.15 (Lagrange's theorem)

If $(S, \oplus)$ is a finite group and $(S', \oplus)$ is a subgroup of $(S, \oplus)$, then $|S'|$ is a divisor of $|S|$. ■

A subgroup S' of a group S is a *proper* subgroup if $S' \neq S$. We'll use the following corollary in the analysis in Section 31.8 of the Miller-Rabin primality test procedure.

Corollary 31.16

If S' is a proper subgroup of a finite group S , then $|S'| \leq |S|/2$. ■

Subgroups generated by an element

Theorem 31.14 affords us a straightforward way to produce a subgroup of a finite group $(S, \oplus)$: choose an element a and take all elements that can be generated from a using the group operation. Specifically, define $a^{(k)}$ for $k \geq 1$ by

$$a^{(k)} = \bigoplus_{i=1}^k a = \underbrace{a \oplus a \oplus \cdots \oplus a}_k .$$

For example, taking $a = 2$ in the group $\mathbb{Z}_6$ yields the sequence

$$a^{(1)}, a^{(2)}, a^{(3)}, \dots = 2, 4, 0, 2, 4, 0, 2, 4, 0, \dots$$

We have $a^{(k)} = ka \bmod n$ in the group $\mathbb{Z}_n$, and $a^{(k)} = a^k \bmod n$ in the group $\mathbb{Z}_n^*$. We define the **subgroup generated by a** , denoted $\langle a \rangle$ or $(\langle a \rangle, \oplus)$, by

$$\langle a \rangle = \{a^{(k)} : k \geq 1\} .$$

We say that a **generates** the subgroup $\langle a \rangle$ or that a is a **generator** of $\langle a \rangle$. Since S is finite, $\langle a \rangle$ is a finite subset of S , possibly including all of S . Since the associativity of $\oplus$ implies

$$a^{(i)} \oplus a^{(j)} = a^{(i+j)} ,$$

$\langle a \rangle$ is closed and therefore, by Theorem 31.14, $\langle a \rangle$ is a subgroup of S . For example, in $\mathbb{Z}_6$, we have

$$\begin{aligned} \langle 0 \rangle &= \{0\} , \\ \langle 1 \rangle &= \{0, 1, 2, 3, 4, 5\} , \\ \langle 2 \rangle &= \{0, 2, 4\} . \end{aligned}$$

Similarly, in $\mathbb{Z}_7^*$, we have

$$\begin{aligned} \langle 1 \rangle &= \{1\} , \\ \langle 2 \rangle &= \{1, 2, 4\} , \\ \langle 3 \rangle &= \{1, 2, 3, 4, 5, 6\} . \end{aligned}$$

The **order** of a (in the group S), denoted $\text{ord}(a)$, is defined as the smallest positive integer t such that $a^{(t)} = e$. (Recall that $e \in S$ is the group identity.)

Theorem 31.17

For any finite group $(S, \oplus)$ and any $a \in S$, the order of a is equal to the size of the subgroup it generates, or $\text{ord}(a) = |\langle a \rangle|$.

Proof Let $t = \text{ord}(a)$. Since $a^{(t)} = e$ and $a^{(t+k)} = a^{(t)} \oplus a^{(k)} = a^{(k)}$ for $k \geq 1$, if $i > t$, then $a^{(i)} = a^{(j)}$ for some $j < i$. Therefore, as we generate elements by a , we see no new elements after $a^{(t)}$. Thus, $\langle a \rangle = \{a^{(1)}, a^{(2)}, \dots, a^{(t)}\}$, and so $|\langle a \rangle| \leq t$. To show that $|\langle a \rangle| \geq t$, we show that each element of the sequence $a^{(1)}, a^{(2)}, \dots, a^{(t)}$ is distinct. Suppose for the purpose of contradiction that $a^{(i)} = a^{(j)}$ for some i and j satisfying $1 \leq i < j \leq t$. Then, $a^{(i+k)} = a^{(j+k)}$ for

$k \geq 0$. But this equation implies that $a^{(i+(t-j))} = a^{(j+(t-j))} = e$, a contradiction, since $i + (t - j) < t$ but t is the least positive value such that $a^{(t)} = e$. Therefore, each element of the sequence $a^{(1)}, a^{(2)}, \dots, a^{(t)}$ is distinct, and $|\langle a \rangle| \geq t$. We conclude that $\text{ord}(a) = |\langle a \rangle|$. ■

Corollary 31.18

The sequence $a^{(1)}, a^{(2)}, \dots$ is periodic with period $t = \text{ord}(a)$, that is, $a^{(i)} = a^{(j)}$ if and only if $i = j \pmod{t}$. ■

Consistent with the above corollary, we define $a^{(0)}$ as e and $a^{(i)}$ as $a^{(i \bmod t)}$, where $t = \text{ord}(a)$, for all integers i .

Corollary 31.19

If $(S, \oplus)$ is a finite group with identity e , then for all $a \in S$,

$$a^{(|S|)} = e.$$

Proof Lagrange's theorem (Theorem 31.15) implies that $\text{ord}(a) \mid |S|$, and so $|S| = 0 \pmod{t}$, where $t = \text{ord}(a)$. Therefore, $a^{(|S|)} = a^{(0)} = e$. ■

Exercises

31.3-1

Draw the group operation tables for the groups $(\mathbb{Z}_4, +_4)$ and $(\mathbb{Z}_5^*, \cdot_5)$. Show that these groups are isomorphic by exhibiting a one-to-one correspondence f between $\mathbb{Z}_4$ and $\mathbb{Z}_5^*$ such that $a + b = c \pmod{4}$ if and only if $f(a) \cdot f(b) = f(c) \pmod{5}$.

31.3-2

List all subgroups of $\mathbb{Z}_9$ and of $\mathbb{Z}_{13}^*$.

31.3-3

Prove Theorem 31.14.

31.3-4

Show that if p is prime and e is a positive integer, then

$$\phi(p^e) = p^{e-1}(p - 1).$$

31.3-5

Show that for any integer $n > 1$ and for any $a \in \mathbb{Z}_n^*$, the function $f_a : \mathbb{Z}_n^* \rightarrow \mathbb{Z}_n^*$ defined by $f_a(x) = ax \bmod n$ is a permutation of $\mathbb{Z}_n^*$.

31.4 Solving modular linear equations

We now consider the problem of finding solutions to the equation

$$ax = b \pmod{n}, \quad (31.26)$$

where $a > 0$ and $n > 0$. This problem has several applications. For example, we'll use it in Section 31.7 as part of the procedure to find keys in the RSA public-key cryptosystem. We assume that a, b , and n are given, and we wish to find all values of x , modulo n , that satisfy equation (31.26). The equation may have zero, one, or more than one such solution.

Let $\langle a \rangle$ denote the subgroup of $\mathbb{Z}_n$ generated by a . Since $\langle a \rangle = \{a^{(x)} : x > 0\} = \{ax \bmod n : x > 0\}$, equation (31.26) has a solution if and only if $[b] \in \langle a \rangle$. Lagrange's theorem (Theorem 31.15) tells us that $|\langle a \rangle|$ must be a divisor of n . The following theorem gives us a precise characterization of $\langle a \rangle$.

Theorem 31.20

For any positive integers a and n , if $d = \gcd(a, n)$, then we have

$$\begin{aligned} \langle a \rangle &= \langle d \rangle \\ &= \{0, d, 2d, \dots, ((n/d) - 1)d\} \end{aligned}$$

in $\mathbb{Z}_n$, and thus

$$|\langle a \rangle| = n/d.$$

Proof We begin by showing that $d \in \langle a \rangle$. Recall that EXTENDED-EUCLID(a, n) returns a triple (d, x, y) such that $ax + ny = d$. Thus, $ax = d \pmod{n}$, so that $d \in \langle a \rangle$. In other words, d is a multiple of a in $\mathbb{Z}_n$.

Since $d \in \langle a \rangle$, it follows that every multiple of d belongs to $\langle a \rangle$, because any multiple of a multiple of a is itself a multiple of a . Thus, $\langle a \rangle$ contains every element in $\{0, d, 2d, \dots, ((n/d) - 1)d\}$. That is, $\langle d \rangle \subseteq \langle a \rangle$.

We now show that $\langle a \rangle \subseteq \langle d \rangle$. If $m \in \langle a \rangle$, then $m = ax \bmod n$ for some integer x , and so $m = ax + ny$ for some integer y . Because $d = \gcd(a, n)$, we know that $d \mid a$ and $d \mid n$, and so $d \mid m$ by equation (31.4). Therefore, $m \in \langle d \rangle$.

Combining these results, we have that $\langle a \rangle = \langle d \rangle$. To see that $|\langle a \rangle| = n/d$, observe that there are exactly n/d multiples of d between 0 and $n - 1$, inclusive. ■

Corollary 31.21

The equation $ax = b \pmod{n}$ is solvable for the unknown x if and only if $d \mid b$, where $d = \gcd(a, n)$.

Proof The equation $ax = b \pmod{n}$ is solvable if and only if $[b] \in \langle a \rangle$, which is the same as saying

$$(b \bmod n) \in \{0, d, 2d, \dots, ((n/d) - 1)d\} ,$$

by Theorem 31.20. If $0 \leq b < n$, then $b \in \langle a \rangle$ if and only if $d \mid b$, since the members of $\langle a \rangle$ are precisely the multiples of d . If $b < 0$ or $b \geq n$, the corollary then follows from the observation that $d \mid b$ if and only if $d \mid (b \bmod n)$, since b and $b \bmod n$ differ by a multiple of n , which is itself a multiple of d . ■

Corollary 31.22

The equation $ax = b \pmod{n}$ either has d distinct solutions modulo n , where $d = \gcd(a, n)$, or it has no solutions.

Proof If $ax = b \pmod{n}$ has a solution, then $b \in \langle a \rangle$. By Theorem 31.17, $\text{ord}(a) = |\langle a \rangle|$, and so Corollary 31.18 and Theorem 31.20 imply that the sequence $ai \bmod n$, for $i = 0, 1, \dots$, is periodic with period $|\langle a \rangle| = n/d$. If $b \in \langle a \rangle$, then b appears exactly d times in the sequence $ai \bmod n$, for $i = 0, 1, \dots, n-1$, since the length- (n/d) block of values $\langle a \rangle$ repeats exactly d times as i increases from 0 to $n-1$. The indices x of the d positions for which $ax \bmod n = b$ are the solutions of the equation $ax = b \pmod{n}$. ■

Theorem 31.23

Let $d = \gcd(a, n)$, and suppose that $d = ax' + ny'$ for some integers x' and y' (for example, as computed by EXTENDED-EUCLID). If $d \mid b$, then the equation $ax = b \pmod{n}$ has as one of its solutions the value x_0 , where

$$x_0 = x'(b/d) \bmod n .$$

Proof We have

$$\begin{aligned} ax_0 &= ax'(b/d) \pmod{n} \\ &= d(b/d) \pmod{n} \quad (\text{because } ax' = d \pmod{n}) \\ &= b \pmod{n} , \end{aligned}$$

and thus x_0 is a solution to $ax = b \pmod{n}$. ■

Theorem 31.24

Suppose that the equation $ax = b \pmod{n}$ is solvable (that is, $d \mid b$, where $d = \gcd(a, n)$) and that x_0 is any solution to this equation. Then, this equation has exactly d distinct solutions, modulo n , given by $x_i = x_0 + i(n/d)$ for $i = 0, 1, \dots, d-1$.

Proof Because $n/d > 0$ and $0 \leq i(n/d) < n$ for $i = 0, 1, \dots, d-1$, the values $x_0, x_1, \dots, x_{d-1}$ are all distinct, modulo n . Since x_0 is a solution of $ax = b \pmod{n}$, we have $ax_0 \bmod n = b \pmod{n}$. Thus, for $i = 0, 1, \dots, d-1$, we have

$$\begin{aligned} ax_i \bmod n &= a(x_0 + in/d) \bmod n \\ &= (ax_0 + ain/d) \bmod n \\ &= ax_0 \bmod n \quad (\text{because } d \mid a \text{ implies that } ain/d \text{ is a multiple of } n) \\ &= b \pmod{n}, \end{aligned}$$

and hence $ax_i = b \pmod{n}$, making x_i a solution, too. By Corollary 31.22, the equation $ax = b \pmod{n}$ has exactly d solutions, so that $x_0, x_1, \dots, x_{d-1}$ must be all of them. ■

We have now developed the mathematics needed to solve the equation $ax = b \pmod{n}$. The procedure MODULAR-LINEAR-EQUATION-SOLVER prints all solutions to this equation. The inputs a and n are arbitrary positive integers, and b is an arbitrary integer.

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MODULAR-LINEAR-EQUATION-SOLVER( $a, b, n$ )
1  ( $d, x', y'$ ) = EXTENDED-EUCLID( $a, n$ )
2  if  $d \mid b$ 
3       $x_0 = x'(b/d) \bmod n$ 
4      for  $i = 0$  to  $d-1$ 
5          print  $(x_0 + i(n/d)) \bmod n$ 
6  else print "no solutions"

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As an example of the operation of MODULAR-LINEAR-EQUATION-SOLVER, consider the equation $14x = 30 \pmod{100}$ (and thus $a = 14$, $b = 30$, and $n = 100$). Calling EXTENDED-EUCLID in line 1 gives $(d, x', y') = (2, -7, 1)$. Since $2 \mid 30$, lines 3–5 execute. Line 3 computes $x_0 = (-7)(15) \bmod 100 = 95$. The **for** loop of lines 4–5 prints the two solutions, 95 and 45.

The procedure MODULAR-LINEAR-EQUATION-SOLVER works as follows. The call to EXTENDED-EUCLID in line 1 returns a triple (d, x', y') such that $d = \gcd(a, n)$ and $d = ax' + ny'$. Therefore, x' is a solution to the equation $ax' = d \pmod{n}$. If d does not divide b , then the equation $ax = b \pmod{n}$ has no solution, by Corollary 31.21. Line 2 checks to see whether $d \mid b$, and if not, line 6 reports that there are no solutions. Otherwise, line 3 computes a solution x_0 to $ax = b \pmod{n}$, as Theorem 31.23 suggests. Given one solution, Theorem 31.24 states that adding multiples of (n/d) , modulo n , yields the other

$d - 1$ solutions. The **for** loop of lines 4–5 prints out all d solutions, beginning with x_0 and spaced n/d apart, modulo n .

MODULAR-LINEAR-EQUATION-SOLVER performs $O(\lg n + \gcd(a, n))$ arithmetic operations, since EXTENDED-EUCLID performs $O(\lg n)$ arithmetic operations, and each iteration of the **for** loop of lines 4–5 performs a constant number of arithmetic operations.

The following corollaries of Theorem 31.24 give specializations of particular interest.

Corollary 31.25

For any $n > 1$, if $\gcd(a, n) = 1$, then the equation $ax = b \pmod{n}$ has a unique solution, modulo n . ■

If $b = 1$, a common case of considerable interest, the x that solves the equation is a *multiplicative inverse* of a , modulo n .

Corollary 31.26

For any $n > 1$, if $\gcd(a, n) = 1$, then the equation $ax = 1 \pmod{n}$ has a unique solution, modulo n . Otherwise, it has no solution. ■

Thanks to Corollary 31.26, the notation $a^{-1} \pmod{n}$ refers to *the* multiplicative inverse of a , modulo n , when a and n are relatively prime. If $\gcd(a, n) = 1$, then the unique solution to the equation $ax = 1 \pmod{n}$ is the integer x returned by EXTENDED-EUCLID, since the equation

$$\gcd(a, n) = 1 = ax + ny$$

implies $ax = 1 \pmod{n}$. Thus, EXTENDED-EUCLID can compute $a^{-1} \pmod{n}$ efficiently.

Exercises

31.4-1

Find all solutions to the equation $35x = 10 \pmod{50}$.

31.4-2

Prove that the equation $ax = ay \pmod{n}$ implies $x = y \pmod{n}$ whenever $\gcd(a, n) = 1$. Show that the condition $\gcd(a, n) = 1$ is necessary by supplying a counterexample with $\gcd(a, n) > 1$.

31.4-3

Consider the following change to line 3 of the procedure MODULAR-LINEAR-EQUATION-SOLVER:

$$3 \quad x_0 = x'(b/d) \bmod (n/d)$$

With this change, will the procedure still work? Explain why or why not.

★ 31.4-4

Let p be prime and $f(x) = (f_0 + f_1x + \cdots + f_tx^t) \pmod{p}$ be a polynomial of degree t , with coefficients f_i drawn from $\mathbb{Z}_p$. We say that $a \in \mathbb{Z}_p$ is a **zero** of f if $f(a) = 0 \pmod{p}$. Prove that if a is a zero of f , then $f(x) = (x - a)g(x) \pmod{p}$ for some polynomial $g(x)$ of degree $t - 1$. Prove by induction on t that if p is prime, then a polynomial $f(x)$ of degree t can have at most t distinct zeros modulo p .

31.5 The Chinese remainder theorem

Around 100 C.E., the Chinese mathematician Sun-Tsü solved the problem of finding those integers x that leave remainders 2, 3, and 2 when divided by 3, 5, and 7 respectively. One such solution is $x = 23$, and all solutions are of the form $23 + 105k$ for arbitrary integers k . The “Chinese remainder theorem” provides a correspondence between a system of equations modulo a set of pairwise relatively prime moduli (for example, 3, 5, and 7) and an equation modulo their product (for example, 105).

The Chinese remainder theorem has two major applications. Let the integer n be factored as $n = n_1n_2 \cdots n_k$, where the factors n_i are pairwise relatively prime. First, the Chinese remainder theorem is a descriptive “structure theorem” that describes the structure of $\mathbb{Z}_n$ as identical to that of the Cartesian product $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_k}$ with componentwise addition and multiplication modulo n_i in the i th component. Second, this description helps in designing efficient algorithms, since working in each of the systems $\mathbb{Z}_{n_i}$ can be more efficient (in terms of bit operations) than working modulo n .

Theorem 31.27 (Chinese remainder theorem)

Let $n = n_1n_2 \cdots n_k$, where the n_i are pairwise relatively prime. Consider the correspondence

$$a \leftrightarrow (a_1, a_2, \dots, a_k), \tag{31.27}$$

where $a \in \mathbb{Z}_n$, $a_i \in \mathbb{Z}_{n_i}$, and